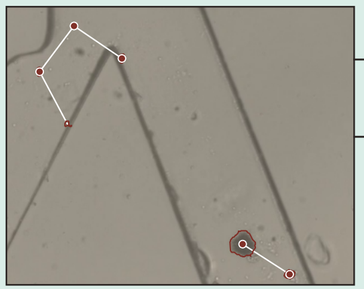
World–Action Model

0.1 s per step · 10-step rollout

Scene inputs



Exp. B · frame 240

Regions · 45-D

Relations · 5-D

PZT + motion · 12-D

Tracks + action log

Encoders

Node MLP · 45 → 64

Edge MLP · 5 → 64

Global MLP · 12 → 64

Processor

Edge MLP

Sum

Node MLP

×3

Global context

Decoder

C

MLP
128 → 8

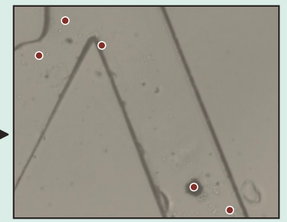
State update

Ensemble

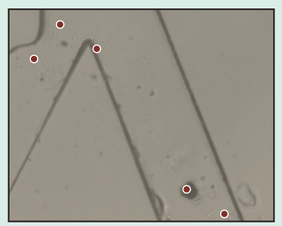
5 members

Mean

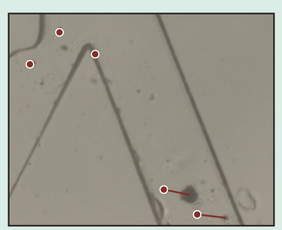
Centroid rollout



Observed t



t + 0.1 s



t + 0.5 s



t + 1.0 s

Message-passing block

×3 distinct weights

Global · 64

Edge

Sender

Receiver

C

Edge update

Linear
256 → 64

SiLU

Linear
64 → 64

SiLU

Linear
64 → 64

+

Sum by receiver

C

Node update

Linear
192 → 64

SiLU

Linear
64 → 64

SiLU

Linear
64 → 64

+

Updated node · 64

Node · 64

C Concatenation + Residual addition

64-D hidden states

Shared weights across time; independent weights across members